TEN-CHANNEL ELECTROACOUSTIC WORK ~~~ FIXED MEDIA

Perpetuum, Interrupted

WRITTEN FOR & DEDICATED TO
SOUND GALLERY · NEW YORK CITY
TEN-SPEAKER ARRAY · MAX/MSP · DAWS

~ SOUND CONCEPT

> The sound design fuses generative patches with live-recorded audio: four Max/MSP systems producing evolving material, interwoven with sounds captured by hand, composed into a fixed ten-channel work diffused across a ten-speaker array surrounding the listener.

~ PROGRAM NOTE

A work written for, and dedicated to, playback on a ten-speaker array at a sound gallery in New York City, in response to AI-generated music and how it dehumanizes and deconstructs the process of creation.

Left to themselves, these patches are cyclical: systems that could run without end. For this piece, I intervened, shaping their endless output into a fixed, linear composition with a beginning, a trajectory, and an ending. That intervention is the point. Where AI generation produces without deciding, this work is generative by the human brain's knowledge of mathematics and music theory, its instinctual creativity, and its willingness to choose.“

Woven through the piece are sounds I captured myself: my own whispering, the furniture in my room, the shaking of tin cans, construction noise, the crossing signals of North Vancouver. Sounds bound to a place, a day, a body. They are a reaction toward what LLMs can never replicate: not the sound, but the witnessing of it.

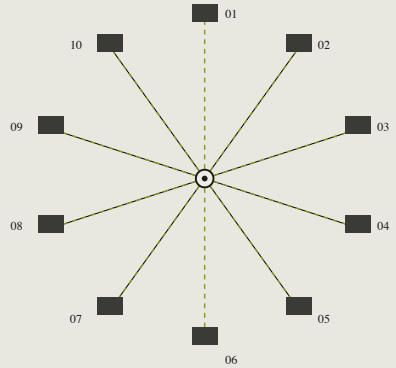
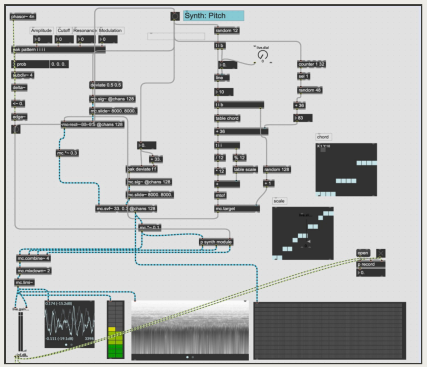


FIG. 1 — PLAN VIEW · TEN-SPEAKER ARRAY, A SOUND GALLERY. LISTENER AT CENTRE; CHANNELS 01–10.

FORMAT	Fixed media, 10-channel (eight full-range, two subwoofer)
SYSTEM	Max/MSP · four generative patches · Logic Pro for linear arrangement · Protools for Mixing
SOURCES	Subtractive synthesis · environmental buffers · foley recordings · construction noise & North Vancouver traffic signals (live-captured)
METHODS	Markov randomization · probabilistic triggering · crossfade & envelope shaping · stretch / pitch-shift / delay

Max/MSP Patches

CYCLICAL SYSTEMS,
SHAPED INTO A LINEAR WORK

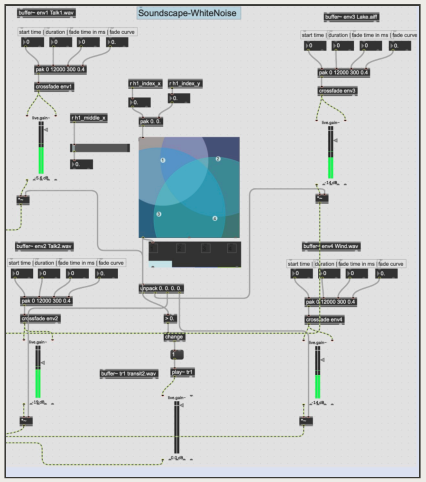


Subtractive synth with pitch, scale & chord controls; outputs visualized with meters and spectral displays.

PATCH 1 · GENERATIVE SYNTHESIZER

This generative synthesizer uses subtractive synthesis to generate sound, acting as a neutral "white canvas" where pitched elements are layered and shaped generatively. Symbolically, it represents the foundation of creativity: a versatile, uncolored base that allows musical ideas to emerge and evolve freely.

live controls: modulation / filtering / amplitude

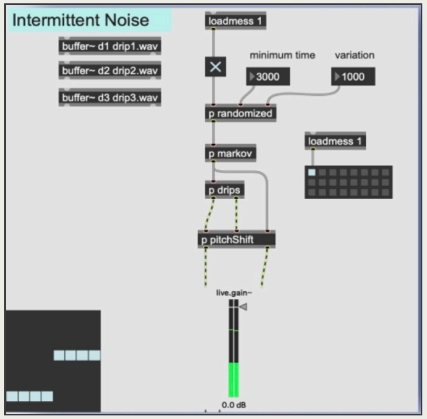


Sliders and numerical inputs adjust duration, fade time and start points.

PATCH 2 · AMBIENT SOUNDSCAPE

A generative soundscape of white noise, symbolically representing real-world objects through the dynamic blending of environmental sounds. It interprets the essence of these objects in an abstract auditory form, bringing their presence into focus.

environmental sound buffers: crossfades / dynamic levels / envelope shaping

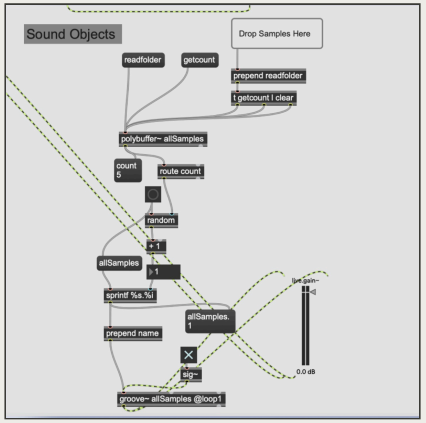


Preset system for recalling states; Markov-based randomization; pitch-shift subpatch modifying signals dynamically.

PATCH 3 · INTERMITTENT NOISE

Intermittent noise appears in irregular bursts rather than continuously, emerging unpredictably with shifting pitch; it represents loneliness and wonder. This third layer adds depth and movement, creating an ethereal, introspective quality that reinforces themes of solitude and curiosity.

drips subpatch: probabilistic triggering of preloaded buffers



Random sample selection and playback from polybuffer~ using random, sprintf and groove~.

PATCH 4 · SOUND OBJECT PLAYER

Inspired by *musique concrète*, the sound object player combines multiple layers of intense everyday noises. Foley-recorded sounds are edited using stretch, pitch shifting, delay and echo, following Schaeffer's concept of reduced listening: removed from their original context, the sounds become abstract textures that evoke the overwhelming sensation of daily anxiety and distraction.

foley sources → stretch / pitch-shift / delay / echo

The Intervention

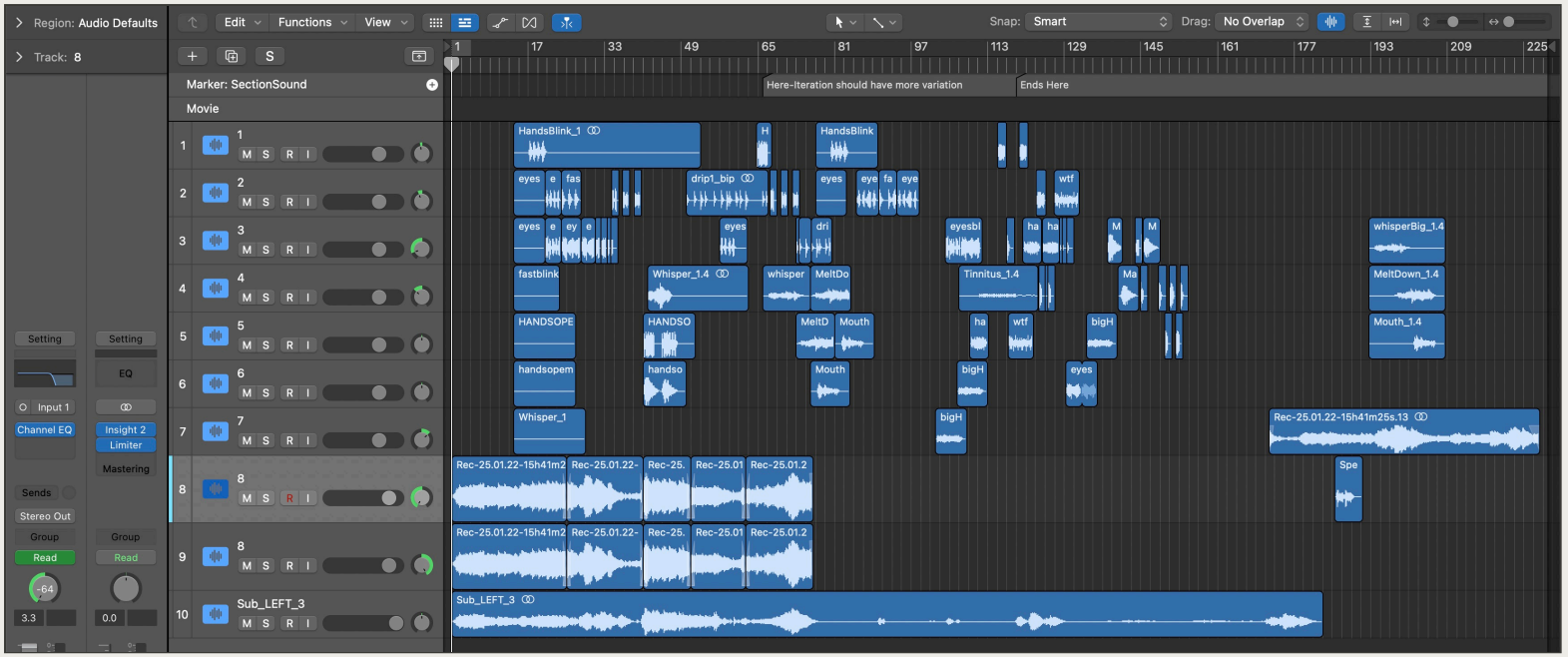


FIG. 2 — LINEAR ARRANGEMENT (LOGIC PRO). THE PATCHES' GENERATIVE OUTPUT, EDITED AND PLACED BY HAND INTO FIXED FORM; THE SESSION MARKER READS "ENDS HERE."

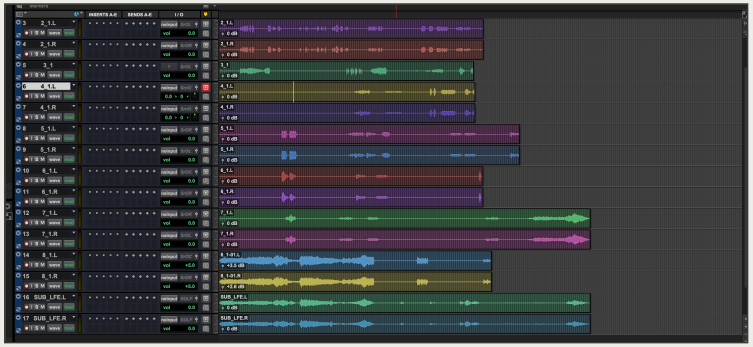


FIG. 3 — TEN-CHANNEL DIFFUSION SESSION (PRO TOOLS). PAIRED STEMS ROUTED TO THE SPEAKER ARRAY, WITH DEDICATED SUB / LFE CHANNELS.



FIG. 4 — I/O CONFIGURATION. BUS ARCHITECTURE MAPPING THE MULTICHANNEL MIX TO PHYSICAL OUTPUTS.